Task 3

Vietnamese Customer Service Callbot

VinSmart Future — AI Internship Program

Issued: June 17, 2026 · Duration: 2 weeks

> **What we're looking for:** We do **not** expect a production-ready callbot. We expect **clear design thinking**, **smooth conversation logic across all 5 categories**, and **honest handling of edge cases and failures**. A bot that gracefully recovers from errors and escalates correctly is better than one that only works on the happy path.

1. Pipeline

Phase	Status	Description
Phase 1 — ASR	**Mandatory**	Capture live microphone input Vietnamese transcript. Any ASR model (PhoWhisper, wav2vec2-vi, Whisper, etc.)
Phase 2 — LLM	**Mandatory**	Process transcript, manage dialogue, generate response. Any local LLM backend (llama.cpp, Ollama, vLLM, HuggingFace, etc.)
Phase 3 — TTS	**Optional (+5 pts)**	Convert text response speech. If not implemented, the bot responds in **text only** — the pipeline is still considered complete.

2. Domain & Categories

The bot operates in the **VinFast customer service** domain, handling 5 categories of inbound calls. The bot must identify the correct category and collect all required fields through natural conversation.

G_1 Cứu hộ
(Roadside Rescue)

Goal	Receive a breakdown report and arrange support (service center dispatch or towing)
Fields	full_name, phone, vehicle_model, license_plate_vin, vehicle_type, current_odo, current_location, city_name, vehicle_condition

G_2 Bảo hành & Sửa chữa
(Warranty & Repair)

Goal	Advise on warranty / repair policy and provide service center information for an appointment
Fields	full_name, owner_phone, vehicle_model, vehicle_usage_type, license_plate_vin, service_center, vehicle_condition

G_3 Đơn hàng
(Order Status & Management)

Goal	Advise on order status and delivery process; handle deposit transfer or order info update
Fields	full_name, order_phone, order_code_dealer, customer_type

G_4 Xe máy – Bảo hành
(Motorbike Warranty)

Goal	Advise on motorbike warranty / repair policy and direct to the nearest service center
Fields	full_name, phone, vehicle_line, license_plate_vin, current_location, vehicle_condition

G_5 Hỗ trợ kỹ thuật từ xa
(Remote Tech Support)

Goal	Guide the customer through a remote fix; book a service center visit if unresolvable
Fields	full_name, phone, license_plate_vin, vehicle_line, current_odo (optional), vehicle_condition_details (incl. software version)

Post-call output — generated from the full conversation transcript at the **end of every call** (not collected during dialogue):

Field	Type	Description
`short_summary`	string	1–2 sentence recap of the call and action taken
`sentimental_analysis`	string	Customer tone throughout the call (e.g. calm / frustrated / urgent)
`emergency`	yes / no	Was there a life-safety or urgent rescue situation?

```
Final output per call: { "category": "G_1", "fields": { ... }, "post_call": { "short_summary": "...", "sentimental_analysis": "urgent", "emergency": "yes" } }
```

3. Exception Handling

The bot must handle all of the following situations. Document your handling strategy for each in the architecture document and demonstrate them in your test scenarios.

Situation	Expected Behaviour
Missing field	Ask only for the missing field — do not re-ask already confirmed information
Customer corrects info	Acknowledge the correction, update the value, continue without repeating confirmed fields
Ambiguous intent	Ask one clarifying question before routing — do not assume a category
Out-of-scope query	Acknowledge scope politely and redirect, or offer to transfer to a human agent
Unclear / garbled input	Ask the customer to confirm unclear values (phone, plate number, etc.) before recording
Emergency detected	Prioritise immediately — provide rescue hotline, skip lower-priority field collection
Stuck after 2+ failed turns	Offer to transfer to a human agent
Customer hangs up mid-call	Output partial JSON with fields collected so far; mark missing fields as null

4. Evaluation Framework

> **Open-ended:** You design your own test scenarios, metrics, and evaluation strategy. There is no fixed benchmark. You will be graded on the **quality and rigour of your evaluation design**, not on hitting a specific score.

Minimum requirements:

- Covers all 5 categories — at least 2 test scenarios per category (10 minimum)
- Includes at least 3 exception handling scenarios from Section 3
- Includes at least one **automated** metric
- Reports failure cases honestly — what the bot got wrong and why
- Measures and reports end-to-end latency per turn

5. Deliverables

#	Deliverable	Description
1	**Architecture document**	Pipeline design, model choices, conversation flow, design decisions and trade-offs
2	**Working bot**	End-to-end runnable pipeline: microphone input ASR LLM text response + extracted fields JSON. TTS voice response if implemented.
3	**Evaluation framework + test scenarios**	Intern-designed test cases, metrics, and evaluation scripts or rubrics. Must include exception handling scenarios.
4	**Evaluation report**	Results across all metrics, failure analysis, latency breakdown, honest assessment of limitations

5	**requirements.txt**	Pinned dependencies. No credentials in code — API keys / tokens via .env only.

6. Scoring Rubric

30	**Pipeline Functionality**	All 5 catego implement
25	**Dialogue Design & Exception Handling**	Conversati Section 3 a
25	**Evaluation Framework**	Covers all 5 Rigour and
20	**Code Quality & Reproducibility**	Clean, read clear setup
+5	**TTS Bonus** — Working speech output. Bot responds in spoken Vietnamese.	

7. Suggested Timeline

Days	Milestone
1 – 3	Architecture design · ASR integration and testing
4 – 7	LLM dialogue covering all 5 categories · Output JSON structure
8 – 10	Exception handling · Edge case testing · TTS (optional)
11 – 14	Evaluation framework · Run tests · Evaluation report · Reproducibility check

Timeline is a suggestion — adjust pacing as needed, as long as all deliverables are submitted by the deadline.